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Systematic review of direct access physiotherapy for musculoskeletal conditions in primary care: consequences for general practitioner workload, resource use, and organisation of care

Matthieu Fischer^{1*} , Emilie Bui¹, Laure Besombes² and Mathilde François^{1,2,3}

Abstract

Background Musculoskeletal disorders (MSDs) are frequent and disabling conditions in primary care and contribute substantially to general practitioners' (GPs) workload. Direct access (DA) to physiotherapy, allowing patients to consult physiotherapists without prior medical referral, has been proposed to streamline care, reduce delays, and optimise resource use. This systematic review synthesized the effects of DA on GPs' practice and the organisation of primary care for adults with MSDs.

Methods Following PRISMA 2020, we systematically searched 12 databases/platforms (MEDLINE/PubMed, Embase, Cochrane Library, CINAHL, PEDro, Web of Science, Scopus, Cairn.info, Érudit, ClinicalTrials.gov) and discovery platforms (ScienceDirect, Google Scholar), plus grey literature, from 15 July 2024 to 4 November 2025 (initial search 15 July–15 October 2024; updated search 1 January 2024–4 November 2025). Eligible studies assessed the consequences of DA to physiotherapy for MSDs in primary care versus GP-referred pathways (referral-based usual care). Two reviewers independently screened, extracted data, as well as appraised quality using the Newcastle–Ottawa Scale (observational studies), ROB 2.0 (RCTs), and graded certainty using GRADE at the study level. Outcomes included GP workload, imaging and medication prescriptions, clinical results, costs, and safety.

Results We included 21 studies (7 RCTs, 14 observational) totaling 90 401 patients (15 017 DA; 75 384 GP-referred). Across 15 studies, DA was associated with 10–30% fewer GP consultations. 14 studies reported reduced imaging and medication use under DA; effects on prescriptions were context-dependent. 14 studies showed non-inferior clinical outcomes, with small early advantages in some RCTs. 13 studies assessed costs, indicating neutral to reduced per-episode costs; two RCT-linked economic analyses were comparable or favorable to DA. No serious adverse events attributable to DA were reported (5 studies). GRADE certainty was moderate in 2, low in 12, and very low in 5 studies,

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limited by observational predominance and small RCT samples. Totals refer to study-level GRADE assessments ($n = 19$), as two economic evaluations were linked to their parent RCTs.

Conclusions In adults with MSDs, DA to physiotherapy appears to lessen GP workload and resource use with clinical outcomes similar to GP-referred pathways and no consistent signal of increased harms. Given methodological limitations, further robust comparative studies are warranted to inform large-scale implementation and policy in primary care.

Trial registration OSF (ID 897ub; registered 15 August 2024; <https://osf.io/897ub/>).

Keywords Direct access, Physiotherapy, Musculoskeletal disorders, Primary care, General practice, Systematic review, Workload, Prescriptions, Imaging, Health services research

Introduction

Musculoskeletal disorders (MSDs) are common and disabling, leading to chronic pain, functional limitations, and loss of autonomy. The World Health Organization (WHO) identifies MSDs as a leading global cause of years lived with disability, affecting about 1.71 billion people ($\approx 22\%$) [1, 2]. In France, MSDs are a major cause of occupational disease and consultation, with substantial economic impact [3, 4]. Management combines pharmacologic and non-pharmacologic treatments, patient education, and continuity of care [5, 6].

General practitioners (GPs) are often the first contact point, who coordinate care pathways, and prescribe investigations/treatments [7–9]. Physiotherapists deliver evidence-based interventions -e.g., therapeutic exercise- to restore function [5, 6]. Growing GP workload and evolving physiotherapist competencies have renewed interest in direct access (DA) to physiotherapy without prior medical prescription. Widely adopted abroad, DA aims to shorten delays, reduce GP burden, and leverage physiotherapists' role; several studies report benefits on pain, function, and patient satisfaction [10–14]. Care under DA appears comparable to physician-referred pathways on clinical outcomes and satisfaction [15–18]. However, effects on GPs and service organization remain under-studied.

Definitions used in this review: Direct Access (DA) refers to patient self-referral to a physiotherapist without prior medical prescription [19]; First Contact Physiotherapy (FCP) denotes a physiotherapist embedded as the initial MSK contact within primary care pathways. The term "triage services" (structured telephone or digital assessment prior to in-person care) is provided here for conceptual orientation; studies involving physiotherapist-led triage were captured through broader DA/FCP search terms (see [Methods](#)).

In France, DA remains limited despite recent pilots/experiments [20, 21]. A historically physician-centered system and legal/cultural/organizational barriers constrain diffusion [21–23]. In the context of strained primary care and the need for safe, accessible organizational innovation, DA warrants systematic evaluation.

Objective

To assess the impact of DA to physiotherapy on general practice for adult MSDs. The primary outcome was GP workload (number/frequency of consultations). Secondary outcomes were prescriptions (medications, imaging), clinical outcomes (pain, function, quality of life), direct/indirect costs, interprofessional collaboration, GP satisfaction/perceptions, and adverse events.

Methods

Study design and reporting framework

This systematic review followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 [24]. In accordance with PRISMA, we prepared outcome-specific Summary of Findings tables (SoF) that collate study-level results per outcome and are provided as Additional Tables S1–S7. The protocol was prospectively registered on the Open Science Framework (OSF; ID 897ub; 15 August 2024, available at <https://osf.io/897ub/>) [25]. The completed PRISMA 2020 checklist is available in Additional file 1.

Search strategy

The initial search was conducted from 15 July 2024 to 15 October 2024. To ensure currency of evidence, an updated search was performed from 1 January 2024 to 4 November 2025, capturing any late-indexed records from the original search period and newly published studies. Searches covered databases (MEDLINE/PubMed, Embase, Cochrane Library, CINAHL, PEDro, Web of Science, Scopus, Cairn.info, Érudit, ClinicalTrials.gov) and discovery platforms (ScienceDirect, Google Scholar). Grey literature was identified through structured searches of Google Scholar (using predefined search equations consistent with the main database strategy), clinical trial registries (ClinicalTrials.gov and WHO ICTRP), and French-language platforms (Cairn.info, Érudit), which index professional and institutional literature not systematically covered by international biomedical databases (see Additional file 2).

Eligibility criteria (PICO framework)

Population: adults (≥ 18 years) with non-specific MSDs (i.e., musculoskeletal conditions without identified specific pathology such as fracture, inflammatory arthritis, or malignancy) managed in primary care; **Intervention:** DA to physiotherapy. **Comparator:** GP-referred pathway (referral-based usual care); **Primary outcome:** GP workload (number/frequency of GP consultations); **Secondary outcomes:** medication/imaging prescriptions, patient-reported outcomes (pain, function, QoL), direct/indirect costs, safety; **Designs included:** RCTs and observational comparative studies (cohort, case-control, quasi-experimental). This combination was chosen to capture both efficacy data from experimental designs (RCTs) and real-world organizational impacts (observational studies) often unavailable in experimental settings. Systematic reviews were excluded from the synthesis and used for context only. Qualitative-only reports, case series, protocols, and non-peer-reviewed documents were excluded. A consolidated PICO framework mapping populations, interventions/comparators, and outcomes across included studies is presented in Additional file 3.

Study selection

Two reviewers (MaF and EB) independently screened titles/abstracts and then full texts using Rayyan for web-based screening and duplicate detection [26]; disagreements were resolved by consensus. For the updated search (1 January 2024–4 November 2025), a pragmatic single-reviewer approach (MaF) was adopted at title/abstract level given the short timeframe. To assess reliability, a random 20% sample ($n = 13/65$) was independently screened by a second reviewer (EB), yielding excellent inter-rater agreement (Cohen's $\kappa = 0.91$). All potentially eligible records were subsequently assessed at full text by both reviewers independently.

Data extraction

Data were extracted using a standardised form adapted from the *Cochrane Handbook for Systematic Reviews of Interventions* (version 6.3) [27], with pre-specified units/scales (e.g., VAS 0–10 or 0–100, ODI 0–100, MSK-HQ 0–56, EQ-5D index) recorded as reported. Extracted items included authors, year, country, design, sample, intervention/comparator, outcomes, measurement tools, sources of funding, declared conflicts of interest, and key quantitative results. Participants were counted once per unique study or cohort to avoid double counting when the same cohort was reported across linked publications (e.g., primary study and linked economic evaluation) or contributed data to multiple outcomes. For transparency, all study-level quantitative results used to populate the SoF are reproduced verbatim in Additional Tables S1–S7, with units and time windows preserved. Two reviewers

(MaF and EB) independently extracted data and reconciled discrepancies by consensus.

Risk of bias and methodological quality

Risk of bias and methodological quality were assessed with Risk of Bias 2.0 (ROB 2.0) for RCTs [28] and the Newcastle–Ottawa Scale (NOS) for observational studies [29]. Ratings were harmonized (low/moderate/high). Risk of bias assessments were performed independently by two reviewers (MaF and EB) with consensus resolution. Detailed study-level quality appraisals are summarized in Additional file 4. A cross-study bias matrix (randomization, selection, performance, detection, attrition, reporting, publication, language, setting, and confounding) is provided in Additional file 5.

Certainty of evidence

Outcome-level certainty was graded with Grading of Recommendations Assessment, Development and Evaluation (GRADE): risk of bias, inconsistency, imprecision, publication bias, indirectness [30]; Given the narrative synthesis without pooled estimates and the heterogeneity of outcomes precluding outcome-level aggregation, GRADE judgements were applied at the study level as inputs to the overall synthesis, consistent with guidance for reviews without meta-analysis. Economic evaluations published separately inherited the parent RCT's certainty rating. Dual independent grading (MaF and EB) with consensus.

Synthesis methods

Given clinical and methodological heterogeneity across designs, settings, and outcomes, we conducted a narrative synthesis. Where available, we report effect measures with 95% CIs and p-values. We did not perform formal small-study or publication-bias tests because no meta-analysis was undertaken; instead, we qualitatively appraised potential reporting/publication biases and reflected these concerns within GRADE. Results from secondary research (systematic reviews) were used for context only and were not recombined with primary studies to avoid double counting.

Ethical considerations

Since only published/public data were used, no ethics approval was required. Protocol/eligibility was preregistered on OSF [25].

Results

Study selection

The combined searches identified 7,264 records (7,239 from databases and 25 from trial registries). After removal of duplicates, 5,003 unique records were screened. The initial search (15 July–15 October 2024)

contributed 6,811 records; the updated search (1 January 2024–4 November 2025) yielded 453 additional records but no further eligible primary studies, likely reflecting evidence base stabilization and publication lag. After screening and eligibility assessment, 21 studies were included. The PRISMA 2020 flow diagram of study selection is presented in Fig. 1. The final sample comprised 7 randomised controlled trials (RCTs) and 14 observational studies. Two economic evaluations published separately were linked to their respective parent RCTs for all quality and certainty assessments to avoid double counting.

Study characteristics

A total of 90,401 adults with musculoskeletal disorders (MSDs) were analyzed (15,017 in direct access [DA] models; 75,384 in GP-referred pathways). This figure represents unique participants per cohort; the higher total of 91,599 sometimes derivable from supplementary tables results from summing sample sizes across linked publications and outcome-specific reports from the same underlying cohorts (see [Methods](#)). Most studies originated from the UK, Sweden, and North America, with one French pilot. MSDs mainly included low back and neck pain, osteoarthritis, and tendinopathies. DA models ranged from fully autonomous first-contact physiotherapy to structured triage-based services.

Funding for 12 of the 17 included reports was provided by public or non-profit institutional grants (e.g., NIHR, Versus Arthritis, regional health services, or university grants). One study was funded by a combination of industry and professional association grants, and four reports did not state a funding source. Regarding competing interests, 12 reports explicitly declared no conflicts of interest. Two reports declared minor editorial conflicts unlikely to influence the outcomes, and three reports did not include a competing interest statement.

Primary outcome—GP workload

Fifteen studies assessed GP workload. Most studies reported a 10–30% reduction in GP consultations under DA vs GP-referral [31–35]; where available, we report 95% CIs/p-values and specify country/setting (e.g., NHS Scotland, UK primary care). UK RCTs confirmed no increase in GP visits [36]. A large Scottish programme estimated 1,739 GP-hours saved per year [32]. Specific quantitative examples: lower imaging rates (7.3% vs 13.6% in an RCT; $p < 0.001$) and fewer GP visits (0.3 vs 1.5 per patient in an RCT-linked economic evaluation) are consistent with these findings [34, 37]. Table 1 summarises the study-level findings for the primary outcome (GP workload) comparing direct access (DA) with GP-referred care; most studies report a reduction in GP consultations under DA, typically around 10–30%.

Secondary outcomes

Prescriptions and imaging

Prescriptions Across 14 studies, DA was generally associated with lower medication use; several cohorts reported 50–85% fewer analgesic prescriptions, including non-steroidal anti-inflammatory drugs (NSAIDs) and opioids [31, 39, 46, 47]. All medication-related outcomes are compiled in Additional file 7 in Table S2 (SoF—Medication use: analgesics/NSAIDs/opioids). Corresponding GRADE ratings and study-level quality (ROB 2.0/NOS) are reported in Additional file 6 and Additional file 4. Some UK cluster contexts showed no reduction or slight increases where triage support was weak [36, 37].

Imaging Comparative studies reported 30–76% lower imaging when DA was structured [32–35]. Pragmatic RCTs confirmed lower X-ray rates (7.3% vs 13.6%, $p < 0.001$) [38]. All imaging outcomes in Additional file 7 in Table S3 (SoF—Imaging: X-ray/advanced imaging). Corresponding GRADE ratings and study-level quality (ROB 2.0/NOS) are reported in Additional file 6 and Additional file 4.

Clinical outcomes

Across 14 studies, DA showed clinical outcomes similar to GP-referred care for pain, function (Musculoskeletal Health Questionnaire [MSK-HQ], Oswestry Disability Index [ODI]), and quality of life (EuroQol 5-Dimension [EQ-5D], 12-Item Short Form Health Survey [SF-12]) [36, 40, 41, 44]. Some RCTs suggested modest early benefits (e.g., -2.2 visual analogue scale [VAS] points at 6 weeks) [41], with no sustained differences mid-term [40, 41, 48]; where available, mean differences with 95% CIs are reported. Study-level clinical results (scales, time-points, mean differences with 95% CI/p when available) are presented in Additional file 7 in Table S4 (SoF—Clinical outcomes).

Costs

Thirteen studies assessed economic outcomes. Economic analyses from observational studies indicated neutral to moderate cost savings under direct access, with per-patient savings of ≈€200–€1,828 [31, 46, 47], with perspective (payer/NHS/societal), time horizon (per episode vs follow-up), and price year reported where available. Two pragmatic RCTs found neutral to modestly reduced total costs, with resource-use components detailed by perspective and price year where available. Fewer GP visits (0.3 vs 1.5, $p < 0.001$) and fewer imaging requests (9% vs 39%) [36, 41] were highlighted. One trial reported a non-significant total-cost difference (–€364, $p = 0.17$) [41]. Itemized cost components and total-cost comparisons by perspective and price year are reported Additional file 7 in Table S5 (SoF -Costs: payer/NHS/societal).

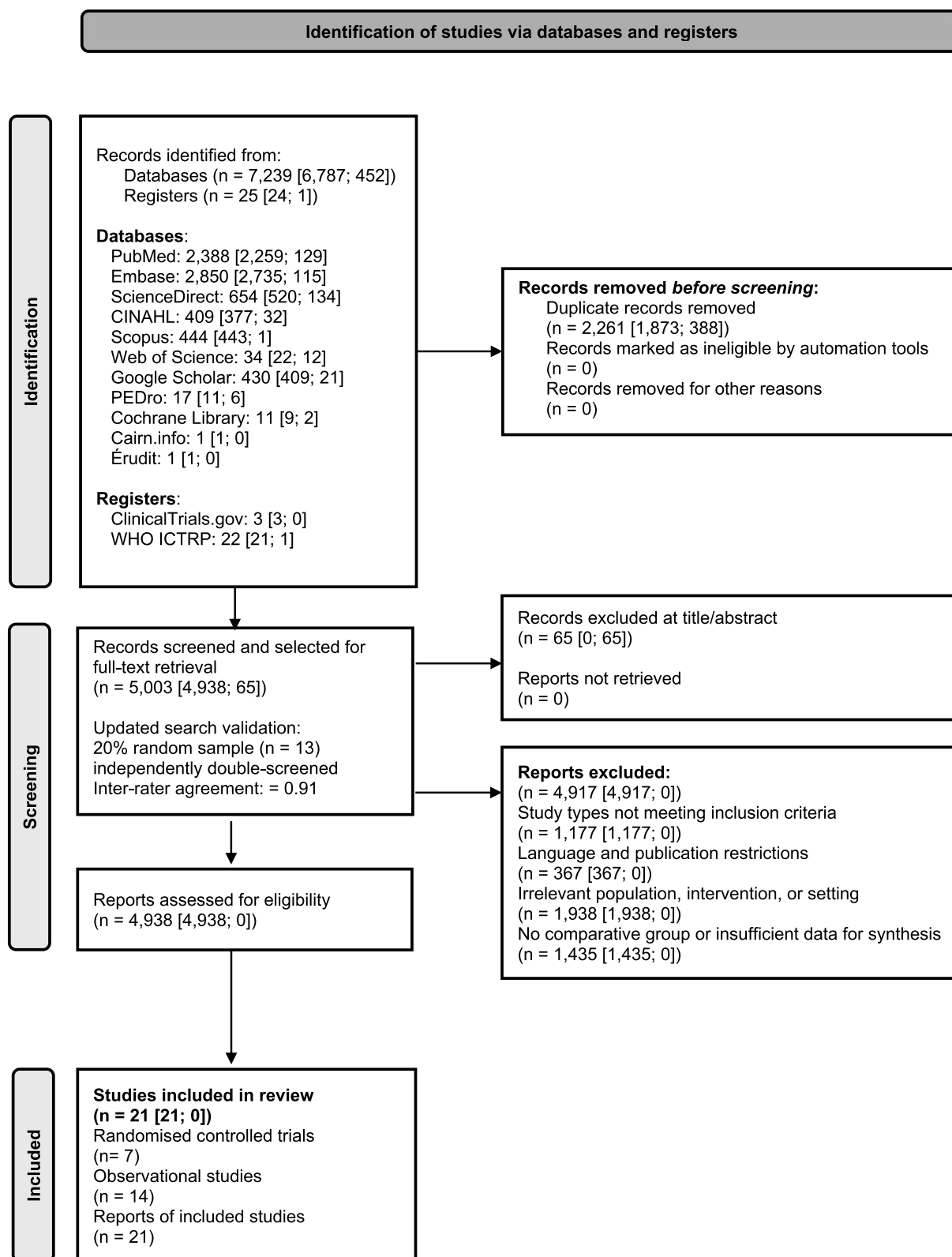


Fig. 1 PRISMA 2020 study selection flow diagram

Table 1 Primary outcome - General practitioner (GP) workload under direct access versus GP-led care

Study (year)	Design & N (window)	Outcome (definition/time)	Arms	Effect (raw)	Statistics (95% CI / p)	Notes (context)
Holdsworth, 2007 [32]	Service evaluation, NR (episode)	GP consultations per episode	SR vs GPR-S vs GPR	0.82 vs 1.70 vs 1.71	$p < 0.001$	UK primary care
Hollinghurst, 2013 [38]	RCT, NR (6 mo)	GP consultations	PhysioDirect vs Usual care	0.87 vs 0.77	NS (overlapping CIs)	Faster access; see S5
Bornhöft, 2015 [39]	Cohort, NR (12 mo)	≥ 2 GP visits	PT-first vs GP-first	OR 0.52 (0.39-0.69)	95% CI shown	≥ 1 visit: OR 0.88 (0.70-1.09)
Bornhöft, 2019 [40]	Cohort, NR (12 mo)	≥ 1 GP visit	PT-first vs GP-first	26% vs 15%	$p = 0.242$	Non-conclusive
Ho-Henriksson, 2022 [41]	Cohort, NR (12 mo)	GP visits per patient	PT-first vs GP-first	0.3 vs 1.5	$p < 0.001$	Cost items concordant (S5)
Ludvigsson, 2012 [42]	Cohort, NR (≤ 3 mo)	GP re-consultation	PT-first vs GP-first	12% vs 48%	$p < 0.001$	85% no GP return (same complaint)
Bishop, 2017 [36]	Pragmatic, NR (12 mo)	GP workload (descriptive)	AD vs Usual care	NR (no comparative figures)	NR	Descriptive only
Bishop, 2021 [37]	Quasi-experiment, NR (2011-2017)	Quarterly GP-MSK consultations per 100	AD sites vs controls	Heterogeneous effect (descriptive)	NR	Practice-level time series
Bastide, 2022 [43]	Local audit, NR (~ 3 mo)	Initial GP visits avoided	AD (CPTS)	-16 visits (≈ 2 GPs)	NR	Simple tally
Stynes, 2021 [44]	Multicentre audit, NR (≤ 3 mo)	GP re-consultation	FCP (no comparator)	20% (10-14%/mo; 10% at 3 mo)	NR	Descriptive
Walsh, 2024 [45]	Pragmatic, NR (3/6 mo)	≥ 1 GP re-consultation	GP vs FCP-St vs FCP-AQ	3 mo: 30.9% vs 15.6% vs 8.8%; 6 mo: 15.6% vs 7.5% vs 5.3%	NR	Mean contacts 0.50 vs 0.23 vs 0.10 (3 mo)

Safety

Across 90,401 patients, no serious adverse events attributable to DA were reported in RCTs [36, 38, 40, 41, 48]; five studies explicitly assessed safety. An observational study documented appropriate red-flag detection within the service evaluated [42]. Safety endpoints (adverse effects, red-flag detection/referral) are summarized in Additional file 7 in Table S6 (SoF—Safety).

Collaboration and GP perceptions

No studies evaluated GP satisfaction, perceived quality of physiotherapy care, or interprofessional collaboration; This gap is documented in Additional file 7 in Table S7 (SoF—Collaboration and GP perceptions), which records the absence of comparative data.

Risk of bias

Among RCTs, most domains were low to moderate risk on ROB 2.0; lack of blinding was inherent, with moderate attrition in two trials [36, 38, 40, 41, 48]. Observational studies showed moderate-high risk (selection, confounding, self-report) [35, 41, 42, 44, 46, 47]. A heatmap summarizing the risk-of-bias domains across included studies is presented in Fig. 2.

Methodological quality

Observational NOS scores ranged 4–6/9 (low-moderate) [32–35, 39, 42, 44, 46, 47]. RCTs were generally of moderate methodological quality but had small sample sizes

and few participating sites, with limited allocation concealment [36, 38, 40, 41, 48]. The distribution of methodological quality levels by appraisal tool (ROB 2.0, NOS) is displayed in Fig. 3.

Certainty of evidence

Using GRADE (study level), 2 studies reached moderate certainty, 12 were low, and 5 very low, often downgraded for risk of bias, imprecision, and inconsistency. The certainty-of-evidence levels according to GRADE and study design are shown in Fig. 4.

Overall synthesis

Evidence suggests DA reduces GP workload and resource use (medications, imaging) while showing clinical outcomes similar to GP-referred pathways and no consistent signal of increased harms. Economic analyses from observational studies indicated neutral to moderate cost savings, though reliance on non-randomised data limits certainty.

Discussion

Main findings and comparison with existing literature and policy

DA for MSDs can reduce GP workload by 10–30%, lower prescriptions and imaging, and achieve clinical outcomes similar to GP-referred care with no consistent signal of increased harms [31, 36, 40, 41, 48]. Economic analyses indicated neutral to moderate cost savings [36, 38, 40,

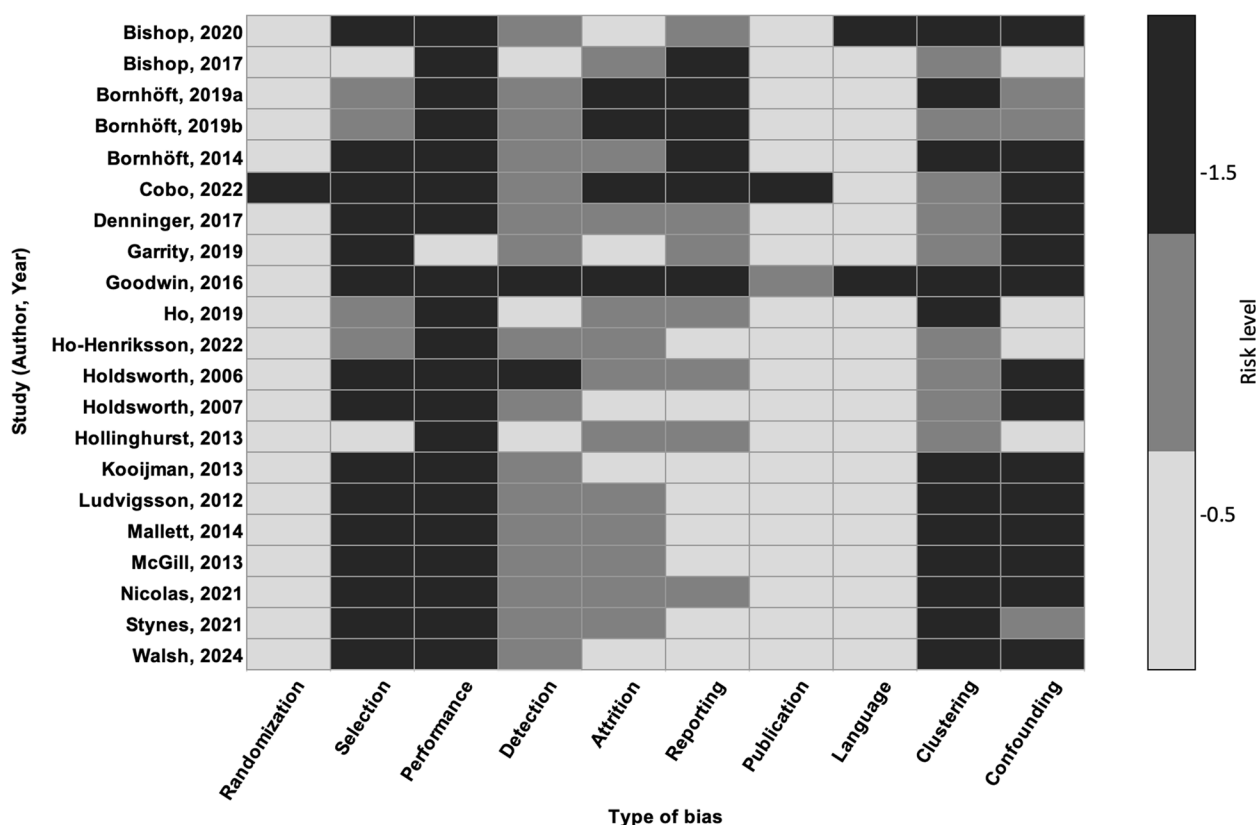


Fig. 2 Risk-of-bias heatmap across included studies

41, 46, 47], though much of it stems from observational evidence. Certainty remains low to moderate overall; only two studies achieved *moderate certainty* according to GRADE (three out of four on the GRADE scale: high, moderate, low, very low).

Our findings align with international trends advocating early physiotherapy access for MSDs to relieve primary care [12, 19, 49–51] and with UK First Contact Physiotherapy (FCP) models [36, 37, 44]. In France, DA is emerging via targeted protocols and recent experimentation [20, 22, 23], with more gradual diffusion than in Anglo-Saxon systems. Limited long-term follow-up precludes firm conclusions on recurrence or delayed medical consumption [36, 37, 40].

Implications for policy across healthcare systems

The findings of this review have differential implications depending on healthcare system architecture. In Beveridgean systems (e.g., UK NHS), direct access has been implemented through First Contact Practitioner (FCP) models embedded within general practice, with physiotherapists serving as first point of contact for MSK complaints. The evidence synthesized here supports the continued expansion of such models, with demonstrated reductions in GP workload and comparable clinical outcomes.

In Bismarckian systems such as France and Germany, where physiotherapy traditionally requires medical prescription and reimbursement is tied to physician referral, implementation faces additional regulatory and cultural barriers. In France, despite recent regulatory developments -including the 2023 Rist law introducing limited direct access [20], the 2024 implementing decree expanding its scope within organised primary care settings [21], and the ministerial order of 25 June 2025 further clarifying eligible conditions and professional requirements [52]-, DA remains limited by: (i) legal requirements for medical prescription; (ii) fee-for-service reimbursement structures not adapted to first-contact physiotherapy; (iii) historical professional boundaries; and (iv) variable physiotherapist training in differential diagnosis and red-flag screening. Our findings suggest that, if these barriers were addressed, direct access could contribute to alleviating the well-documented pressures on French general practice, particularly in underserved areas.

In liberal healthcare systems (e.g., USA), where DA legislation varies by state, the evidence supports unrestricted access models, with studies consistently showing lower costs and comparable outcomes. However, implementation considerations differ regarding insurance coverage and out-of-pocket costs for patients.

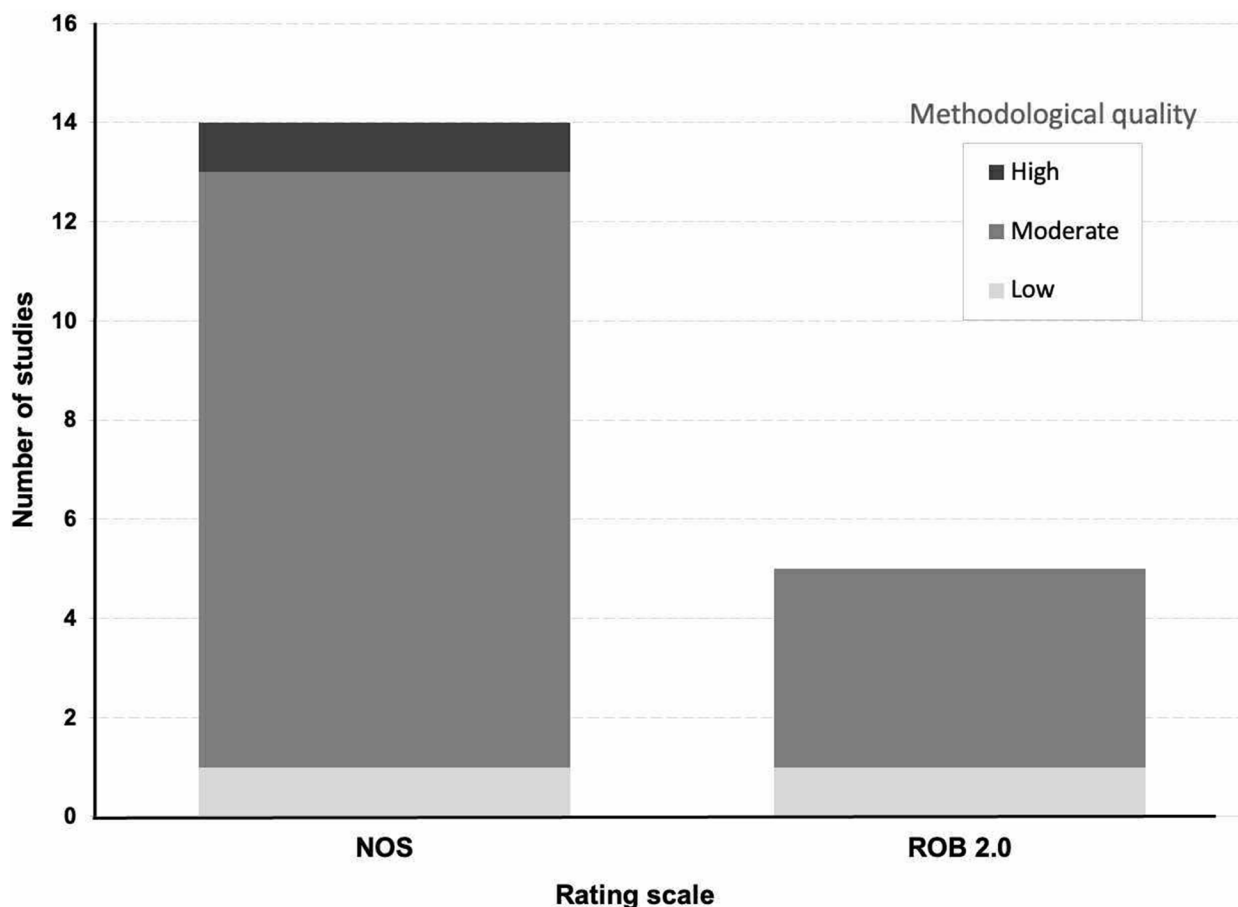


Fig. 3 Methodological quality distribution (ROB 2.0 / NOS)

Across all systems, successful implementation requires: clear triage protocols and red-flag screening competencies; explicit thresholds for medical referral; robust communication channels between physiotherapists and physicians; context-specific reimbursement models; and ongoing monitoring of workload, outcomes, and safety.

Interpreting certainty of evidence for policy decisions

The predominance of low and very low GRADE certainty ratings across outcomes reflects the observational nature of most included studies, a limitation inherent to health services research where randomization is often unfeasible or ethically problematic. This "low certainty" rating should not be interpreted as "poor quality evidence" but rather as appropriate epistemic humility given the study designs available.

Importantly, low certainty does not imply that policy action should be deferred. GRADE guidance explicitly acknowledges that strong recommendations can be made despite low certainty evidence when: (i) the direction of effect is consistent across studies; (ii) the magnitude of potential benefits substantially outweighs potential harms; and (iii) the consequences of inaction

are significant. All three conditions are arguably met here: seven RCTs and fourteen observational studies consistently point toward reduced GP workload with no consistent signal of increased harms; potential harms appear minimal; and primary care systems in many countries face unsustainable workload pressures.

The two studies achieving moderate certainty for selected outcomes (GP workload proxies, costs) provide an anchor of higher-confidence evidence, while the convergent findings from multiple lower-certainty studies across diverse healthcare systems provide corroborating support. For decision-makers, this triangulation of consistent findings offers a robust directional signal for policy, even if precise effect magnitudes remain uncertain. Future pragmatic trials and well-designed prospective cohorts could strengthen certainty, but should not be prerequisites for cautious implementation with appropriate monitoring.

Strengths and limitations

Strengths include PRISMA-guided methods, multi-database search with prospective update to November 2025, double independent screening/appraisal, and use of

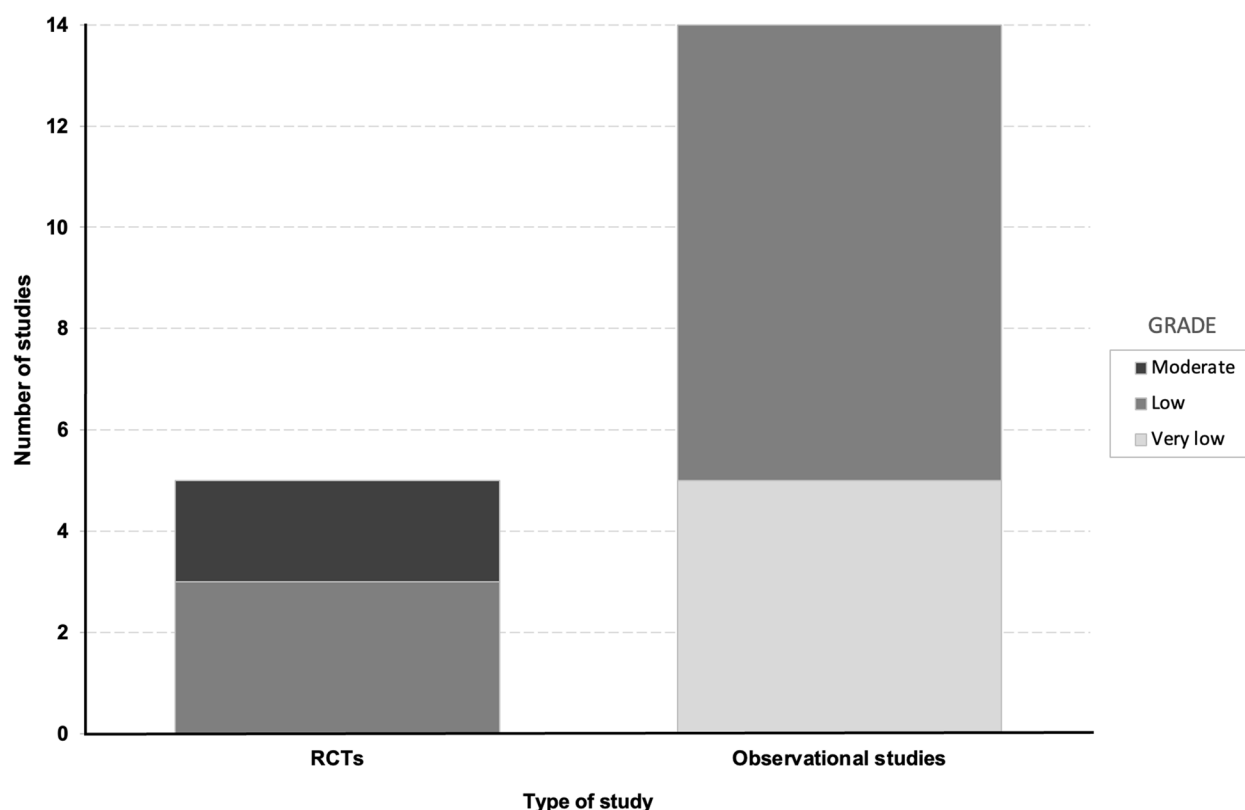


Fig. 4 GRADE certainty by study design

validated tools (ROB 2.0, NOS, & GRADE). The breadth of health system contexts and organisational models covered also strengthens the external validity of the findings, particularly for health services research and policy-oriented interpretation. *Limitations* include heterogeneity of DA/FCP/triage models and outcomes, predominance of observational designs, small RCT samples, and short follow-up, which prevented a robust meta-analysis; potential publication/reporting biases are considered within the GRADE judgements.

Implications for practice, organisation, and future research

DA appears safe and may alleviate GP consultation workload, supporting more rational imaging and prescribing [32–36, 38]. For France, scaling would require structured triage/protocols [20, 22, 23], interoperable tools, explicit collaboration frameworks [15, 16], and training in differential diagnosis for physiotherapists [49]; evaluation within community health centres could monitor real-world impact [21, 53, 54]. Beyond clinical safety, implementation should be viewed as a reconfiguration of care pathways, with implications for professional roles, accountability, and continuity of care in primary care teams. Embedding DA within coordinated organisational models may therefore be as critical as regulatory change to ensure sustainable adoption.

Priorities include pragmatic RCTs and prospective cohorts in regulated systems, with adequate samples, long-term outcomes, and full economic evaluations [36, 37, 40, 41]; and studies on GP satisfaction, collaboration, and patient acceptability [15, 16]. Future research should also adopt mixed-methods designs to better capture contextual factors influencing implementation, including organisational readiness and interprofessional dynamics. Such approaches are essential to inform policy decisions in systems where DA is introduced incrementally rather than through wholesale reform.

Conclusions

In adults with MSDs, direct access to physiotherapy appears to reduce GP workload and resource use (medications, imaging) with clinical outcomes similar to GP-referred pathways and no consistent signal of increased harms. Given the predominance of observational designs and small pragmatic trials in the included evidence, future research should prioritize adequately powered RCTs and long-term economic evaluations in regulated healthcare systems, alongside studies of GP acceptability and interprofessional collaboration.

Abbreviations

CPTS Communautés professionnelles territoriales de santé (territorial professional health communities)

DA	Direct access
EQ-5D	EuroQol-5 Dimension
FCP	First Contact Physiotherapy
GRADE	Grading of Recommendations Assessment, Development and Evaluation
GP	General practitioner
MSD(s)	Musculoskeletal disorder(s)
MSK-HQ	Musculoskeletal Health Questionnaire
N	Total number of participants analyzed in the study (not a priori calculated sample size)
MSP	Maisons de santé pluriprofessionnelles (multidisciplinary primary care centers)
NOS	Newcastle-Ottawa Scale
NSAIDs	Non-steroidal anti-inflammatory drugs
ODI	Oswestry Disability Index
OSF	Open Science Framework
RCT	Randomised controlled trial
ROB 2.0	Risk of Bias 2.0
SF-12	12-Item Short Form Health Survey
VAS	Visual analogue scale
WHO	World Health Organization

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12875-026-03186-9>.

Additional file 1: PRISMA 2020 checklist (completed).

Additional file 2: Search strategies (all sources, 2004–2025; initial search 15 Jul–15 Oct 2024; updated search 1 Jan 2024–4 Nov 2025).

Additional file 3: PICO framework across included studies.

Additional file 4: Methodological quality appraisals (ROB 2.0 / NOS).

Additional file 5: Bias matrix across studies.

Additional file 6: GRADE certainty (ratings & rationales).

Additional file 7: Summary of Findings (SoF) tables S1–S7.

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Use of generative AI

Generative AI (ChatGPT, OpenAI) was used only for language editing and formatting. The authors reviewed and are fully responsible for the content. No AI-generated images or data were used.

Authors' contributions

MaF (Matthieu Fischer): Conceptualization; Methodology; Investigation; Data curation; Formal analysis; Writing—original draft; Supervision. *EB (Emilie Bui):* Investigation; Formal analysis; Writing—review and editing. *LB (Laure Besombes):* Methodology; Supervision; Writing—review and editing. *MFr (Mathilde François):* Methodology; Supervision; Writing—review and editing.

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Data availability

The dataset(s) supporting the conclusions of this article are available in the Open Science Framework (OSF) repository, [<https://osf.io/897ub/1>] (<https://osf.io/897ub/>). Additional materials (full search strategies, screening logs, extraction forms, and risk-of-bias tables) are provided as Additional files.

Declarations

Ethics approval and consent to participate

Not applicable. This study is a systematic review of previously published data and did not involve new collection of human participants' data or animal experiments.

Consent for publication

Not applicable. The manuscript does not contain data from any identifiable individual person.

Competing interests

The authors declare no competing interests.

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